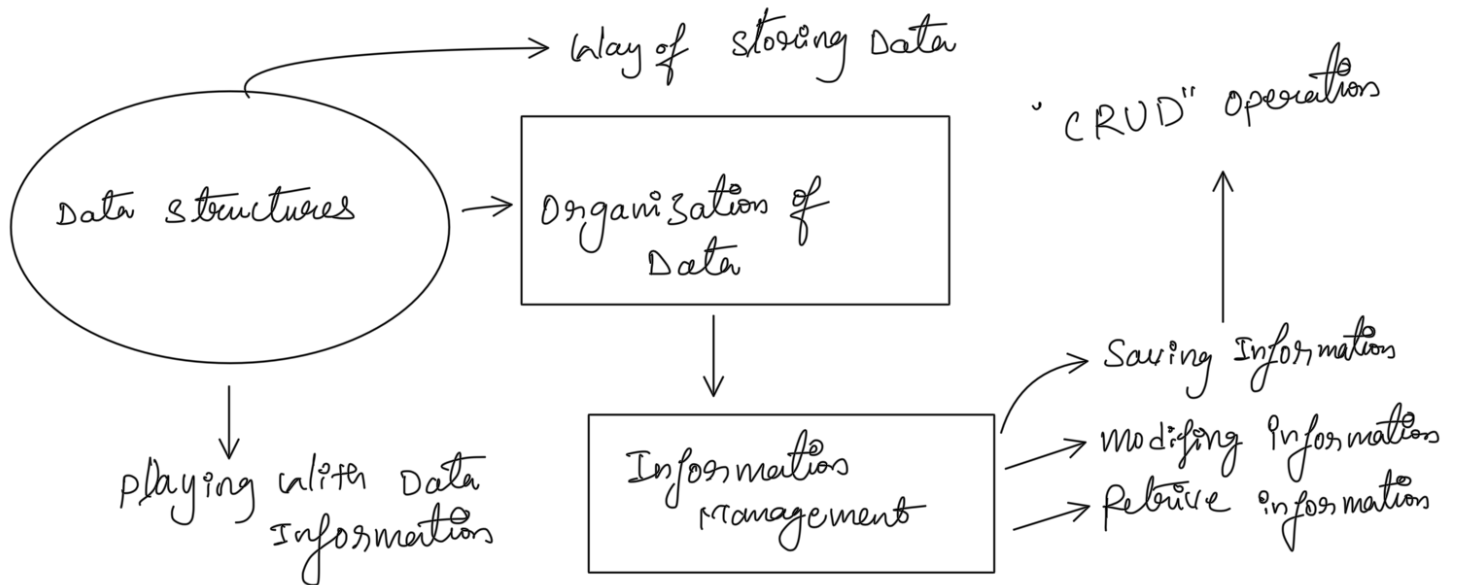


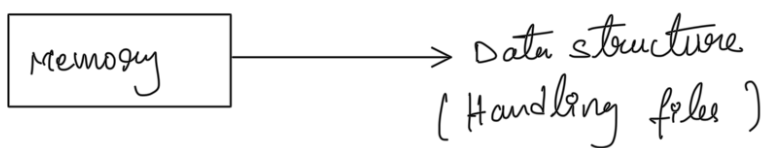
INTRODUCTION TO DATA STRUCTURES

Day 31
10/08/2025

- What are Data structures
- Why we need Data structure
- Why we study time and Space complexity?
- Types (Linear, Non Linear)



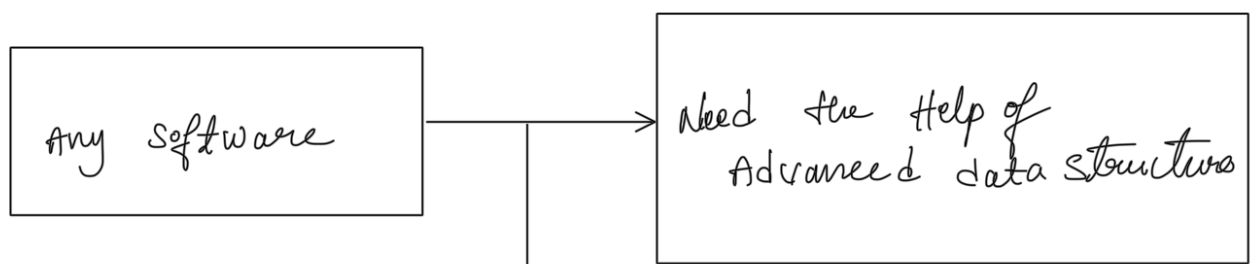
Almost problems in the computer science are about Information Management



Computer science is all about solving problems related to Managing data. (These problems are solved by Data structures)

Why Data structures?

In computer science either we are sending the information, receiving the information, storing information. All these operations need to be managed in a systematic way.



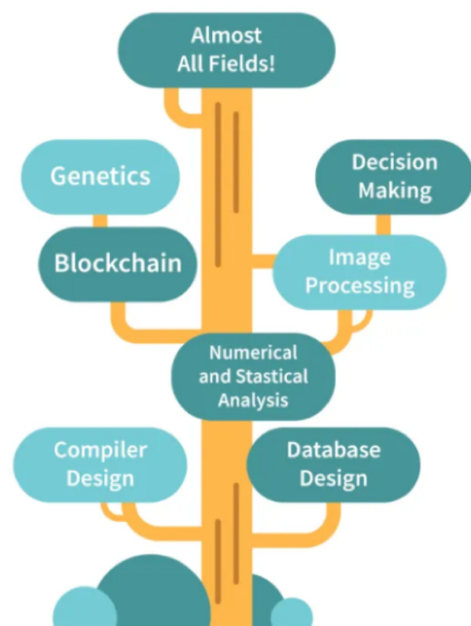
Because the Information in Any software is complex
So we need Help of Advanced data structures

Why data structures is important for computer engineering?

- * You can not do anything without data on world . If you want to proceed add operator, you should have at least two number.
- * Computer sciences deal with storing, organizing and retrieving effectively of data.
- * Computer programmer should get data from user or another sources and use them.
- * for this purpose you have to use data structure for every software program or system.

3

Applications of Data Structures



Data Structures

To create a well optimised software's we have to be good in understanding of Data structures.

Why Time & Space complexity ?

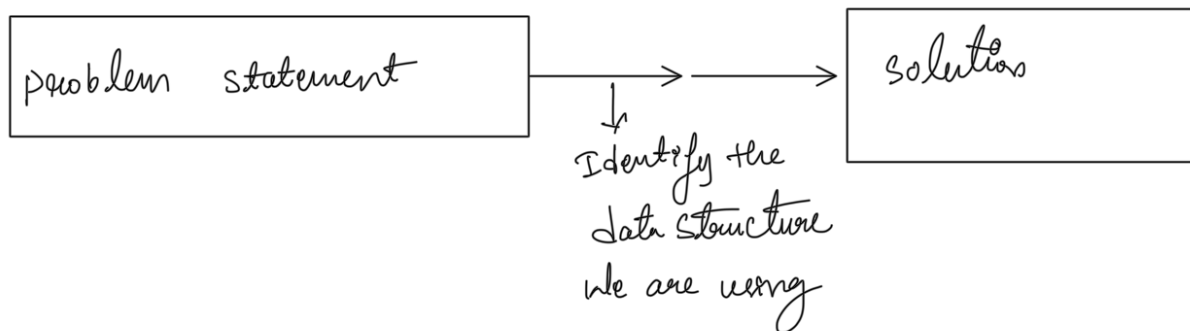
To Execute a Program

Time Complexity	Space Complexity
Calculates the time required	Estimates the space (memory) required
Time is counted for all statement	Memory space is counted for all variables, inputs and outputs.
The size of the input data is the primary	Primary determinant is the auxiliary variable

determinant	size
Deals with the computational time with the change in the size of the input	Deals with how much (extra) space would be required with a change in the input size.

To Build faster and time efficient softwares, understanding of Time & space complexity of program is must

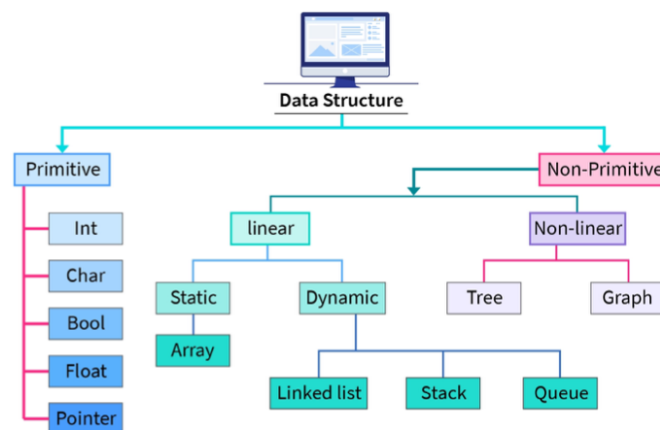
- Types of Data structures (about way of storing data)



Each data structures are capable of solving different problems in its own way

We have to choose the efficient way.

So different data structures are used to solve different problems.



Types